

Ioniser audit and maintenance checklist

If you walked onto your production floor right now, these are the essential steps to verify:

- **Measure decay time at the true point of use.** Not near the ioniser, not at the audit station, rather at the exact handling surface.
- **Validate ion balance under actual airflow conditions.** Ensure conveyors, fans, and tools are running during measurement.
- **Inspect emitter pins.** Look for dust, oxidation, or residue; replace or clean as required.
- **Re-check placement and distance.** Ideal spacing is usually 30.48cm×45.72cm (12-18 inches) from the target zone.
- **Review maintenance records.** Identify trends balance drift, decaying airflow, and worsening decay times.
- **Test under worst-case environmental conditions.** Peak humidity changes, maximum operator movement, or concurrent machine operations.

intermittent failures. Consider a typical production scenario: a connector or cable assembly quietly accumulates 150V, while the adjacent device can tolerate only 85V before a CDM failure occurs. No visible spark or audible event takes place, yet a latent field failure has already been introduced.

This is where ioniser behaviour becomes critical. An ioniser with excellent decay time but poor balance can inject unwanted charge into a product. Conversely, an ioniser with perfect balance but poor placement may neutralise so slowly that a CDM event slips through regardless.

Such oversights recur across manufacturing floors:

- Ionisers mounted too high→Neutralisation slows dramatically by the time ions reach the work surface.
- Emitter contamination from dust, flux vapour or oxide→Balance drifts from a healthy $\pm 10V$ to an unsafe $\pm 80V$ in just weeks.
- Airflow turbulence created by conveyors, fans or heating, ventilation, and air-conditioning (HVAC)→neutralisation becomes inconsistent across the work zone.

And here's the key point: none of these are equipment failures; they are application failures. They come from how ionisers are installed, understood, and monitored, not from faults inside the hardware itself.

Ionisers age and their ageing is invisible

Ionisers do not remain factory-fresh indefinitely, and their degradation is

rarely visible until it causes damage. Over time, emitter pins foul and oxidise; airflow decreases as filters load up; ion balance drifts as electrodes wear; and even cleanrooms accumulate insulating surfaces that alter charge behaviour.

To keep pace with increasingly sensitive devices, best practices now treat ionisers as active, continuously monitored tools rather than set-and-forget equipment. Lines handling ultra-low-CDM components increasingly require weekly critical parameter checks, while most assembly operations benefit from monthly decay-time and balance measurements.

Calibration is shifting towards point-of-use testing rather than isolated benchtop verification, ensuring measurements reflect real operating conditions. Many facilities are also adopting self-cleaning emitters or closed-loop feedback ionisers that automatically maintain balance. With device voltage tolerances now shrinking to 20–40V, even small performance drifts can place products at immediate risk.

Why $\pm 35V$ balance is no longer safe

The long-established $\pm 35V$ ANSI/ESD guideline was developed for older technologies with CDM thresholds in the hundreds of volts, levels considered robust by the standards of that era. Modern ICs, however, can fail at just 20-40V, rendering $\pm 35V$ far too high and even $\pm 25V$

potentially unsafe.

As a result, many fabrication facilities now target $\pm 10V$ or lower for sensitive devices. Ioniser limits are being customised for the most vulnerable components, units are positioned closer to actual handling zones, and decay and balance checks have shifted from quarterly to weekly routines. Performance changes are logged to support predictive maintenance.

In this environment, ionisers are no longer passive equipment; they are active, continuously monitored elements of the production process.

The real issue?

As device geometries shrink and CDM thresholds collapse, decay time and offset voltage have moved from obscure specifications to frontline safeguards in ESD control. Many unexplained failures are not caused by worn equipment. Still, by assumption, that compliance automatically equates to safety, that ionisers require infrequent measurement, or that legacy CDM limits still apply to modern components.

The solution is straightforward. Ioniser behaviour must be managed with the same rigour applied to humidity control, thermal limits, or solder-reflow profiles. Ionisers do not protect products simply by being present; effective protection depends on continuous understanding, measurement, and active management of performance.

In modern electronics manufacturing, ionisers are no longer background players. They are critical guardians of device integrity. When components fail quietly despite passing audits, decay time and offset voltage deserve close scrutiny. In a world where a few volts can determine the difference between a pass and a field failure, complacency is no longer an option. **EFY**

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